

Date	03 October 2026
Team ID	6ab8f5efc23107aaaa7325ec
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture

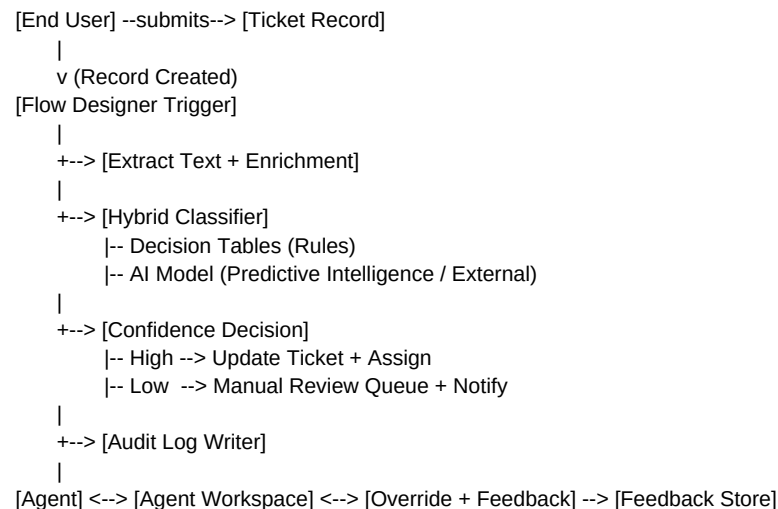
1. Architecture Overview

The solution is an event-driven, low-code architecture centred on Flow Designer. It sits inside the existing ITSM platform and adds an intelligent classification layer without requiring new user interfaces or custom servers.

Logical Layers:

- Presentation Layer: Service Portal (End User) + Agent Workspace (Agent) – unchanged.
- Orchestration Layer: Flow Designer (the core of the solution).
- Intelligence Layer: Decision Tables (Rules) + AI Classification Model (Predictive Intelligence / AI Builder / External LLM).
- Data Layer: Ticket/Incident table + Classification Audit Log + Feedback store + CMDB (read).
- Integration Layer: Optional REST / Integration Hub calls to external AI services.

2. High-Level Component Diagram (Textual)



3. Key Architectural Decisions

- **Event-driven:** Classification runs only when a ticket is created (or updated if re-classification is needed). No polling.
- **Hybrid Intelligence:** Rules handle high-volume known patterns with perfect explainability; AI handles the long tail.
- **Platform-native first:** Prefer Flow Designer + Decision Tables + Predictive Intelligence over external custom code.
- **Human-in-the-loop:** Low-confidence path and agent override guarantee that automation never becomes a black box.
- **Audit by design:** Every decision is logged with inputs, outputs, confidence and model version.

- **Stateless flow steps:** Easy horizontal scaling; platform manages capacity.

4. Data Flow Summary

- Ticket created → Flow triggered → Text extracted → Classification executed → Ticket fields updated → Assignment performed → Audit record written → (optional) Agent notified.
- Agent feedback → stored against original prediction → later used for rule tuning or model retraining.

5. Non-Functional Architecture Considerations

Concern	Approach
Performance	Target < 5 s end-to-end. Caching of frequent patterns. Async where possible.
Reliability	Flow retries + fallback to Manual Review. No ticket left unassigned.
Security	Least-privilege service account, ACLs, encrypted transit, full audit log.
Scalability	Platform-managed; flow is stateless. Supports growth to high daily volumes.
Maintainability	Rules in Decision Tables (editable by process owners). Versioned AI models.

6. Deployment View

Everything runs inside the existing SaaS platform instance. No additional servers, containers or network components are required for the core solution. Optional external AI endpoint is called over HTTPS via Integration Hub / REST.

7. Architecture Principles Followed

- Keep it simple – prefer configuration over code
- Fail safe – always have a human path
- Observe everything – full auditability
- Improve continuously – feedback loop designed in
- Respect the platform – upgrade-safe, native capabilities first